CIFAR-10 Image Classification using Convolutional Neural Networks

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1 Introduction

Image classification is used in computer vision where models assign labels to visual data. Convolutional Neural Networks (CNNs) have become the best approach as they learn hierarchical feature representations from raw pixel inputs (Zhao et al., 2024).

CNN architectures have convolutional layers, pooling operations, activation functions and fully connected layers to transform low-level features into better quality representations. This allows CNNs to outperform traditional machine learning approaches that rely on manual feature extraction (Zhao et al., 2024).

CNNs are still efficient and effective even after newer developed models particularly in applications where computational resources are limited (Korkmaz, 2025). This project aims to develop, evaluate and optimise CNN models for the CIFAR-10 dataset. the main focus will be understanding how preprocessing, architecture and optimisation strategies effect performance.

2 Data Preprocessing and Augmentation

The CIFAR-10 dataset has 60,000 images across 10 classes with a resolution of 32×32 pixels. Due to the low resolution and visual similarity between some classes the datastet is a big classification problem.

Pixel values were normalised to the range [0,1] to improve numerical stability and training efficiency. Sample images were visualised before training to better understand class variability. This showed that certain classes such as cats and dogs, share similar visual feature so there was a need to use techniques that improve generalisation.

Data augmentation was applied using transformations such as horizontal flipping, rotation and translation. Augmentation was used to increase dataset diversity and allow the model to learn invariant representations instead of memorising specific patterns.

This approach is supported by Dogan et al. (2024) who show that augmentation improves CNN performance by increasing robustness to variations in image structure. In this project augmentation was important due to the dataset's limited resolution and high intra-class similarity.

Augmentation also allows the model to learn spatially invariant features and not need fixed object positions. This is useful for CIFAR-10 where objects have different positions. The model becomes less sensitive to minor transformations and better able to generalise to unseen data.

3 Model Architecture

A baseline CNN architecture was designed using multiple convolutional and pooling layers followed by fully connected layers. Convolutional layers extract spatial features, while pooling layers reduce dimensionality and computational complexity.

The design of convolutional and pooling layers helps model performance as these operations control how features are extracted and compressed (Zafar

et al., 2024). Increasing the number of filters enhances feature representation but also increases computational cost.

An improved model was developed using batch normalisation and dropout. Batch normalisation stabilises training by normalising intermediate activations and dropout reduces overfitting by randomly disabling neurons during training.

These techniques are used for improving CNN performance and generalisation (Darwish, 2024). In this project they were introduced to improve training stability and reduce overfitting in the more complex model.

The architectural design also has a balance between model capacity and efficiency. Deeper networks can capture more complex patterns but they also increase the risk of overfitting and require more resources. The model architecture used has enough representational power without introducing unnecessary complexity.

4 Training and Evaluation

The models were trained using the Adam optimiser and categorical cross-entropy loss function. Adam was chosen for an adaptive learning rate and efficient convergence properties.

Training was conducted using a batch size of 64 to balance computational efficiency with gradient updates. Early stopping was applied to prevent overfitting by stopping training when validation performance ceased to improve.

The choice of optimiser and training configuration plays a key role in convergence behaviour. The Adam optimiser was selected for its efficiency and also for its ability to adapt learning rates during training which helps maintain stable optimisation across different layers of the network.

4.1 Experimental Setup

The models were implemented using TensorFlow/Keras. A learning rate of 0.001 was used with the Adam optimiser. The baseline model was trained for 10 epochs and the improved and tuned models were trained for up to 15 epochs with early stopping applied based on validation loss. A validation split of 20% was used during training to monitor generalisation performance.

Model performance was evaluated using accuracy, loss curves, confusion matrices and classification metrics for high-level and detailed insights into model behaviour.

4.2 Baseline Model Performance

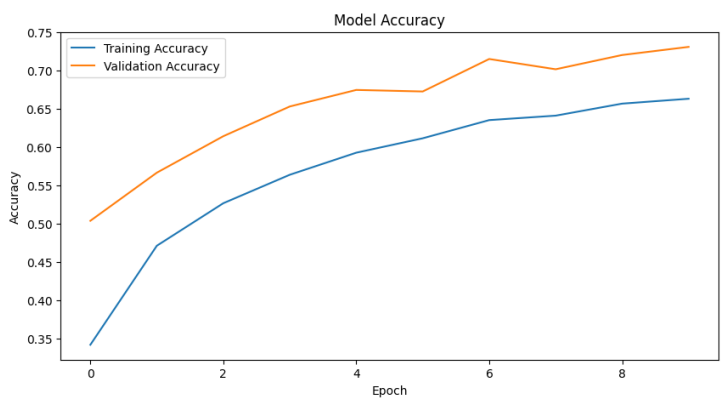

Figure 1: Training and validation accuracy of the baseline model.

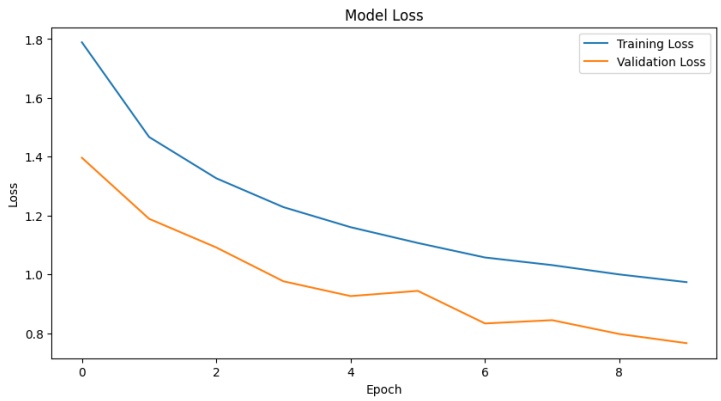

Figure 2: Training and validation loss of the baseline model.

Figure 1 and Figure 2 show that the baseline model has steady improvement in both training and validation performance. The small gap between the curves suggests good generalisation but overall accuracy is limited due to the model's simple architecture.

This result suggests that the baseline model is capable of learning general patterns but it has limited depth and lack of regularisation which reduce its ability to capture more complex feature relationships within the dataset.

4.3 Improved Model Performance

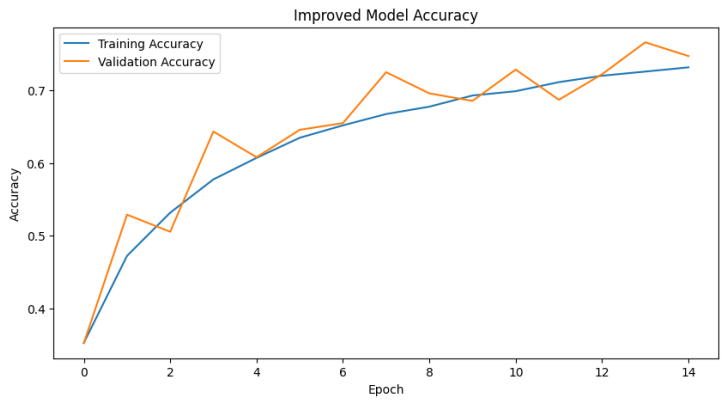

Figure 3: Training and validation accuracy of the improved model.

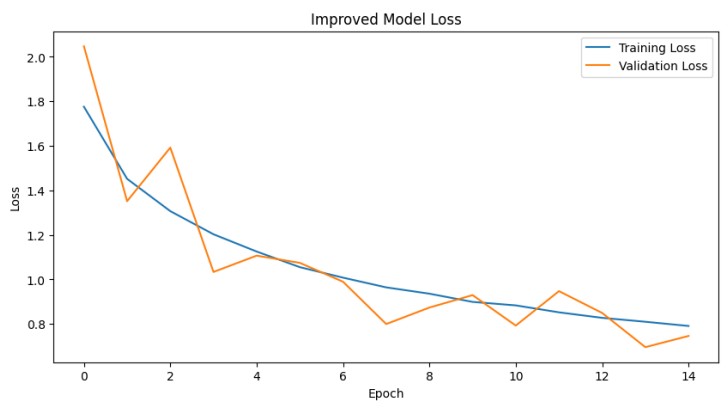

Figure 4: Training and validation loss of the improved model.

Figure 3 and Figure 4 show that the improved model achieves higher validation accuracy than the baseline model. This improvement can be is because of the inclusion of batch normalisation and dropout which improve training stability and reduce overfitting.

Divergence between training and validation loss in epochs indicates overfitting. This shows a trade-off between model capacity and generalisation

where increased complexity allows the model to capture more detailed patterns but also increases memorisation.

The improved architecture improves feature extraction but the increased model capacity should be controlled. Without proper regularisation or data diversity the model prioritises training accuracy over generalisation performance.

4.4 Confusion Matrix Analysis

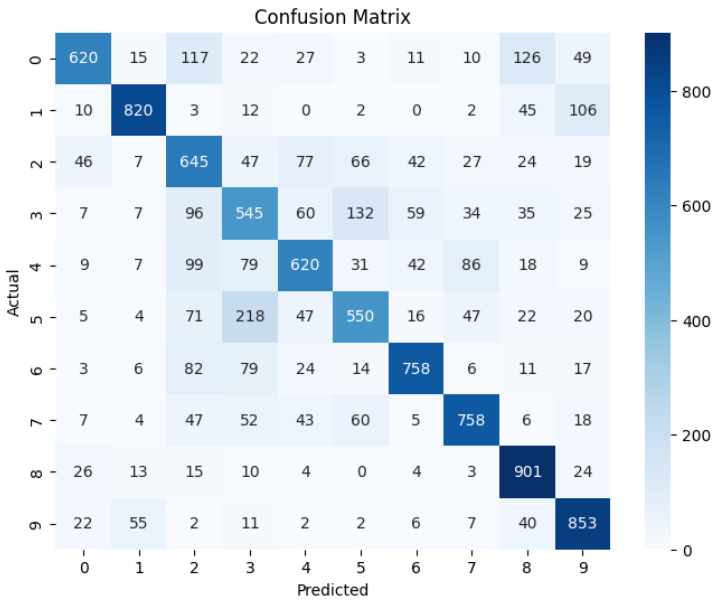

Figure 5: Confusion matrix showing classification performance across CIFAR-10 classes.

Figure 5 the confusion matrix shows strong performance for classes with distinct features such as automobiles and ships. Misclassification occurs between visually similar classes such as cats and dogs and deer and horses.

This limitation is due to the low resolution of the dataset and the similarity of feature representations between classes. This is common in findings

that CNN performance is influenced by feature separability and dataset complexity (Darwish, 2024).

This misclassification shows the importance of feature discrimination in CNNs. When classes have similar textures or shapes the learned feature representations overlap reducing classification confidence and increasing error rates.

5 Model Improvement and Optimisation

Data augmentation improved generalisation by increasing training data variability. Batch normalisation stabilised training and dropout reduced overfitting through regularisation.

These techniques improved model performance and the quality of feature learning. Augmentation introduced variability, batch normalisation improved optimisation behaviour and dropout reduced reliance on individual features.

These two techniques show performance improvements don't depend on increasing model complexity, they depend on improving how effectively the model learns from the available data.

5.1 Hyperparameter Tuning and Analysis

Hyperparameter tuning was done by varying learning rate, batch size and dropout rate. Multiple configurations were tested but the tuned model did not outperform the improved model.

This can be explained by the limited search space and relatively short training duration which restricted the ability to identify best configurations. The improved model may already have been close to an the best configuration leading to diminishing returns from further tuning.

These findings highlight that improvements often have a greater impact than hyperparameter tuning alone (Darwish, 2024).

This also suggests that tuning isolated parameters may have limited impact when the overall learning behaviour is already well-structured. Greater improvements may require changes to feature extraction or model design rather than optimisation settings alone.

6 Model Comparison and Final Selection

The performance of the models is summarised as follows:

- Baseline Model: 72.9%
- Improved Model: 75.5%
- Tuned Model: 75.1%

The improved model was selected as the final model as it achieved the highest accuracy while maintaining strong generalisation. This demonstrates that improvements had a greater impact than tuning alone.

This comparison reinforces the importance of evaluating both model accuracy and generalisation behaviour when selecting a final model instead of just using performance metrics.

7 Limitations

Several limitations were identified. The CIFAR-10 dataset's low resolution makes it difficult to distinguish visually similar classes. The hyperparameter tuning process was limited and the models were trained for a small number of epochs. More advanced techniques such as transfer learning were not explored.

These limitations show how model performance depends on data quality and model design so improvements in either area could lead to more significant performance gains.

8 Conclusion

This project shows that CNN models can effectively classify CIFAR-10 images. The improved model achieved the best performance using batch normalisation, dropout and data augmentation.

The results emphasise that architectural design and regularisation techniques are more impactful than hyperparameter tuning alone. Improvements in feature representation were more important for performance than small parameter adjustments.

The study shows that effective model development requires a combination of preprocessing, carefully designed architectures and well structured evaluation strategies.

9 Future Work

Future work could explore transfer learning, larger hyperparameter search spaces and higher-resolution datasets to further improve performance.

10 Code Availability

The code can be accessed online through the links:

- GitHub Repository
- Google Colab Notebook

References

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